



Does executive function affect children's peer relationships more than emotion understanding? A Longitudinal study based on latent growth model

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ABSTRACT

Many researchers argue that emotion understanding and executive function can help people establish and maintain positive peer relationships, but the existing empirical evidence supporting this assertion in preschool children remains inconclusive. This study aimed to investigate the impact of emotion understanding and executive function on children's peer relationships (specifically, acceptance and rejection) by utilizing a conditional latent growth model. Longitudinal data were collected in three time points within two years from a sample of 120 kindergarten children ($M_{age} = 46.63$ months, with 70 boys) in China. Our findings revealed a significant linear increase in both emotion understanding and executive function over time. The conditional latent growth model further showed that the initial level of emotion understanding and executive function positively predicted later peer acceptance, while the initial level of executive function negatively predicted peer rejection, even after accounting for covariates such as age, gender, family socioeconomic status, and vocabulary. Furthermore, children with faster growth rate of executive function were less likely to experience peer rejection. Intriguingly, the development of emotion understanding did not yield a discernible impact on peer relationships. Overall, these findings suggest that executive function plays a more crucial role than emotion understanding in facilitating positive peer relationships among Chinese preschool children, thereby highlighting the significance of executive function's developmental trends in social interactions.

1. Introduction

As the nursery rhyme goes, “Mom and Dad go to work, and I’m off to kindergarten.” In Chinese education settings, it is expected for children aged 3 and above to separate from their parents and engage in peer group activities at kindergarten. Undeniably, peers hold a significant and influential position in children's lives, alongside their parents. Extensive research has emphasized the function of peer relationships in children's social adjustment, emotional well-being, academic performance, and school adaptation, both at present and in the future (Eggum-Wilkens et al., 2014; Gardner & Gerdes, 2015; Greenman et al., 2009; Hernández et al., 2016). The formation of peer relationships is closely related with children's cognitive abilities and emotional skills (Brownell & Hazen, 2010; Sendil & Erden, 2014), particularly their capacity to control thoughts and actions (e.g., executive function) and

their skill to understand others' emotions (e.g., emotion understanding) (Garcia-Andres et al., 2010; Hughes et al., 1998; Nicholas et al., 2012; Slaughter et al., 2015; Wang et al., 2021). Gaining insight into the development of peer relationships in children is of great significance. Nevertheless, there has been limited research exploring the connection between emotion understanding, executive function, and the formation of peer relationships in preschool-aged children. Specifically, emotion understanding and executive function are commonly recognized as crucial skills for grasping social situations and responding effectively to them (Fruchtman-Steinbok et al., 2016; Hendry et al., 2016; Schurz et al., 2020), yet their precise contributions to the formation of peer relationships remain uncertain.

The preschool years mark a pivotal stage in the development of peer relationships and interactions, as children begin to expand their social networks beyond family members (Ortega et al., 2009). Peers are the

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primary social group with whom children interact during their time in kindergarten, apart from their family and parents. Within this group, some children are accepted by their peers, while others may be ignored or actively rejected, influencing the structure of peer groups and determining each child's position within the group (Fink et al., 2014; Hay et al., 2010). Existing research has mainly focused on a single dimension of peer relationship (e.g., social preference or social popularity), but relying on one dimension may not fully capture the complexity of social relationships (Andrews, 2020; Rubin et al., 2006; Zhang et al., 2014). To illustrate, even children who are generally accepted by their peers can still experience instances of rejection (Bukowski et al., 2010). In light of the complex nature of peer groups, gaining a better understanding of children's peer relationship may come from simultaneously considering two key indicators: peer acceptance and peer rejection (Babcock et al., 2014; Bukowski et al., 2010; Lecce et al., 2020). Furthermore, peer relationships are the external manifestation of socialization for various emotional and cognitive outcomes (Rubin et al., 2006). It is worth noting that the most studies on peer relationships have relied on cross-sectional designs (Moriguchi, 2014; Wang et al., 2021), making it challenging to establish clear causality and stability between variables, such as the influence of cognitive abilities on peer relationships. The preschool stage encompasses both the initial establishment and maintenance of relationships within a new peer group (Fink et al., 2014; Wang et al., 2021). The acquisition and consolidation of emotion understanding and executive function are also developmental processes that require time, experience, and maturation for these skills to be maximized in practical peer relationships (Kuhnert et al., 2017). Therefore, using a longitudinal tracking study, our goal is to investigate how changes in emotion understanding and executive function over time influence peer relationships for acceptance and rejection, within the framework of Chinese preschool environments.

1.1. Emotion understanding and peer relationships

Emotion understanding, as a facet of social cognition, encompasses the capacity to predict and explain emotion in terms of causes, consequences, and contextual determinants (Ana & Tenenbaum, 2013; Carpendale & Lewis, 2006). In general, the ability to understand one's own emotions, particularly the ability to gain insight and respond to emotional cues in a socially appropriate way, is a vital skill essential for successful peer interactions (Imuta et al., 2016; Kim & Cicchetti, 2010; Longobardi et al., 2015). On one hand, children with advanced emotion understanding may have an advantage in establishing peer relationships as they are better equipped to recognize and understand the emotions of their peers (Gulay & Onder, 2013; Hughes & Leekam, 2010; Slaughter et al., 2002). That is, emotion understanding enables children to distinguish and respond to social-emotional situations more effectively. On the other hand, children with a higher level of emotion understanding are more likely to be well-liked by their peers, which in turn provides a social context that supports the development of their emotional skills (Hernández et al., 2017), including their emotion understanding. Thus, children who possess greater emotion understanding are more likely to be accepted by their peers within a group. Previous studies have also demonstrated that difficulties in emotion understanding lead to problematic peer relationships, particularly an increased risk of peer rejection (Goodman & Southam-Gerow, 2010; Herd & Kim-Spoon, 2021). For example, children with ADHD often face challenges in accurately recognizing and identifying the emotions of their peers (Fonseca et al., 2009), which may explain their higher likelihood of experiencing peer rejection than typically developing children (Forner et al., 2017; Gardner & Gerdes, 2015). Furthermore, research suggests that temperament is a personal characteristic that influences children's peer relationships (Acar et al., 2015; Acar et al., 2019; Rudasill et al., 2013), and is related to the emotion knowledge of relationship development (Szweczyk-Sokolowski et al., 2005). Specifically, children with a difficult temperament (e.g., high reactivity and low

inhibitory control) often exhibit heightened negative emotions and are prone to using maladaptive behavioral strategies (e.g., aggression or withdrawal) when facing stress within peer relationships. Children in this situation are at a higher risk of peer rejection (Acar et al., 2019). In contrast, children with a pleasant temperament have more effective emotional skills that facilitate their popularity during peer interactions.

Although prior research has highlighted the crucial role of children's emotion understanding as a fundamental skill in their peer relationships (Denham et al., 1990; Hernández et al., 2017), it remains unclear how the development of emotion understanding over time impact the formation of these relationships. In particular, the initial level of emotion understanding during early childhood is likely to restrict both individual interaction skills and the establishment of relationships with peers (Denham & Brown, 2010; Malti & Noam, 2016), but the effect of growth and change on peer relationships is blank. Therefore, considering the temporal dynamics, it is essential to examine the developmental effects of emotion understanding on peer relationships, particularly regarding peer acceptance and rejection.

1.2. Executive function and peer relationships

It is evidence that peer relationships are also affected by children's cognitive ability during development. Specifically, cognitive functions influence both the content of peer interactions, and the interactive and communicative skills (Hay et al., 2010; Sendil & Erden, 2014). Executive function has been defined as a range of higher-order cognitive functions serving to execute appropriate actions and inhibit inappropriate actions for the attainment of specific goals (Diamond, 2013; Garon et al., 2008). This includes key components such as inhibition control, cognitive flexibility, and working memory (Davidson et al., 2006; Hendry et al., 2016; Miyake & Friedman, 2012). The role of executive function in social adaptation is fundamental, as it assists children in handling challenges, integrating information, and adjusting to dynamic social environments (Caporaso et al., 2019; Lecce et al., 2017). These abilities establish a close association between executive function and peer relationships during the preschool years. On one hand, successful peer interactions require friendlier and more self-regulated behaviors (Denham & Brown, 2010; Ladd, 2005). On the other hand, difficulties in executive function can lead to negative peer experiences (Diamantopoulou et al., 2007; Holmes et al., 2016; Lengua, 2003). Consequently, children's executive function may serve as a strong predictor of group peer relationships (Moriguchi, 2014).

Previous research has demonstrated that deficiencies in inhibitory control are associated with increased externalizing problems in 1st and 2nd grade children (Riggs et al., 2004), and a reduced ability to regulate impulsive behaviors (Jahromi & Stifter, 2008). Difficulties in cognitive flexibility may result in preschool children being unable to modify their initial strategies or response choices in light of negative consequence (Wang et al., 2021). Moreover, reduced working memory has shown associated with social problems and deficits in conflict resolution skills (Kofler et al., 2011; McQuade et al., 2013; Ramos et al., 2019). Most of these studies have focused on one components of executive function, providing evidence for predicting the impact of executive function on peer relationships, but without drawing definitive conclusions about the direct effect. A Longitudinal study by Holmes et al. (2016) examined the developmental links between executive function and peer-related problems, revealing that children who exhibited lower executive function at 4.5 years old experience more difficulties with peers at ages 6 and 9, when compared to their peers (Holmes et al., 2016). Garcia-Andres et al. (2010) found that children ages 7-8 with higher executive function were more popular among their peers, while those with lower executive function were more likely to face rejection (Garcia-Andres et al., 2010). However, these works provide limited information on the developmental relations between executive function and peer relationships during early childhood, and whether they affect peer acceptance and rejection differently. Hence, this study aims to further investigate the

potentially important role of executive function in predicting peer relationships among preschool children, offering a broader perspective from a developmental standpoint.

1.3. Emotion understanding, executive function and peer relationships

The social information processing model (Crick & Dodge, 1994; Turiel, 1998) posits that children's understanding and interpretation of situations impact their behaviors, and that both cognitive and emotional processes contribute to the development of social competence in children (Lemerise & Arsenio, 2003). It is evident that for children to establish positive peer relationships, they need not only to understand and express emotions, but also require more appropriate and self-regulated behaviors during their interactions with peers. Despite the clear concurrent associations among executive function, emotion understanding and peer relationships (Hay et al., 2010), few studies have included both emotion understanding and executive function as independent variables to explore their differential effects on peer relationships during the preschool years. Researchers have suggested that the link between various cognitive skills and peer relationships is likely to change over time (Wang et al., 2021). Individual differences in emotion understanding among children aged 2 to 10 have an important and lasting influence on later social relationships from both theoretical and practical perspective (Slaughter et al., 2015). Nonetheless, previous studies have underscored the heightened impact of executive function on peer relationships during early childhood (de Wilde et al., 2016; Harms et al., 2014). For instance, at the age of 4.5, executive function exhibits a better predictor of peer-related problems at ages 6 and 9, in addition to emotional skills (Holmes et al., 2016). Furthermore, the development of executive function may facilitate important cognitive skills for peer relationships such as theory of mind, communicative behaviors, and moral competence (Devine et al., 2016; Kochanska et al., 1997; Wang et al., 2016). For example, the ability to understand and express one's own thoughts and emotion requires a certain level of executive function development (Austin et al., 2014). Together, the roots of childhood individual differences in peer relationships can be traced to social skills such as executive function and emotion understanding demonstrated in early interactions with peers. Considering the potential distinct roles of emotional and cognitive skills in social interactions, thus, the current study included both emotion understanding and executive function to investigate the varying effects of their temporal dynamics on peer relationships during early childhood.

1.4. The temporal dynamics of longitudinal data in emotion understanding and executive function

Given the anticipated dynamic nature of peer relationships and ability development in early childhood, our interest lies in exploring variations throughout the kindergarten years to understand how the temporal dynamics of executive function and emotion understanding influence the formation of peer relationships. The whole picture of temporal dynamics can be revealed by its initial level and growth rate (Hamaker et al., 2015). As a developmental starting point, the initial level not only reflects early mental ability development but also serves as a significant predictor of subsequent levels of mental ability development. The growth rate depicts the increasing and decreasing trend of individual mental ability in the process of development and reflects the direction and potential for changes in children's mental ability (Szumski et al., 2019). At the same time, the relationship between the initial level and the growth rate can exhibit the opposite patterns of high starting point with fast growth or low starting point with fast growth (Brown et al., 2005). Therefore, when exploring the development of emotional understanding and executive function, we should consider both the initial level and growth rate simultaneously, thereby elucidating the influence of these two indicators on peer relationships (McMaster et al., 2005). On one hand, children with a high initial level

of emotion understanding and executive function exhibit better interaction performance in new social environments, making them more likely to be accepted by their peers within a group (Caporaso et al., 2019; Hernández et al., 2017; Garcia-Andres et al., 2010). On the other hand, the formation of peer relations is also affected by the growth rate of emotion understanding and executive function. For instance, significant improvement in a child's theory of mind during development may attract more playmates (Fink et al., 2014). However, to date, most longitudinal studies in this field have only focused on explain whether the levels of emotion understanding and executive function are related to children's peer relationships or whether they can predict peer outcomes (e.g. Holmes et al., 2016; Kuhnert et al., 2017). These studies fail to directly address the impact of increasing or decreasing trends in these abilities over time on peer outcomes, such as acceptance or rejection. The unique influence of growth rates on peer relationships is grossly overlooked, resulting in the developmental links remaining at a static level and significantly diminishing the application of information related to dynamic changes provided by longitudinal data. To this end, we utilized latent growth models to analyze longitudinal data in order to draw more accurate conclusions about individual trajectories of developmental processes among executive function, emotion understanding, and peer relationships.

1.5. Peer Relationships in Chinese Preschool Settings

Different cultural communities provide distinct developmental pathways for children in diverse cultural contexts (Bronfenbrenner & Morris, 2007). In Chinese culture, children are taught to control their inappropriate behavior and strive to get “Small red flower” (A representational reward) in kindergarten (Chen, 2010; Chen et al., 2005). Parents place more emphasis on their children's performance on school tasks (e.g., drawing, counting, etc.) and hope that their children can befriend academically successful peers. In addition, Chinese parents generally demand self-control (also known as executive function) in their children's daily (Chao & Tseng, 2002). In contrast, social skills such as emotion understanding appear to be undervalued in Chinese preschool settings. Nevertheless, the goals of education in Chinese preschools also include helping children learn social skills to maintain harmonious relationships with peers. Despite this, there is limited research within the context of Chinese preschool settings to explore which has a greater impact on peer relationships in terms of executive function or emotion understanding.

Furthermore, there are important practical implications in investigating the formation of peer relationships during the preschool stage, prior to children transitioning to formal schooling. For example, in kindergarten, it is crucial to focus on developing children's executive function to address peer-related issues. Given that children with more advanced executive function tend to be more popular among their peers, it is highly probable that this trend will persist into formal schooling—a pivotal phase for nurturing social and academic success trajectories (Leerkes et al., 2008; Wagner et al., 2021). Also, exploration of the temporal dynamics of executive function and emotion understanding can offer early educators' evidence-based guidance for deciding when to implement different interventions to enhance children's peer relationships.

1.6. The present study

Given the current state of research, there is a clear need for further investigation into the influence of executive function and emotion understanding on the formation of children's peer relationships, specifically in the context of Chinese preschool settings. Additionally, the differential role of the growth rate of executive function and emotion understanding over time in predicting peer outcomes, in addition to static initial states, should be examined. More longitudinal data on the temporal dynamics of executive function and emotion understanding is

also necessary. Thus, the primary purpose of this study is to investigate the influence of emotion understanding and executive function on the formation of peer relationships in children aged 4 to 6 years through a longitudinal study. To achieve this, we employed a conditional latent growth model to analyze the initial levels and growth rates of emotion understanding and executive function. Given the complex nature of peer group (Bukowski et al., 2010), the study focused on two indicators—peer acceptance and peer rejection—as measures of children's peer relationships. Data were collected at three time points within two years from a sample of 120 children, with measurements of both executive function and emotion understanding taken at each time point. Peer acceptance and rejection were measured through peer nomination at time 3.

Previous research has indicated that family socioeconomic status (SES), child gender, and age are associated with peer relationships (Li et al., 2011; David & Murphy, 2007) and executive function (Hackman et al., 2015; Lawson & Farah, 2017). Therefore, in this study, we included the child's gender, age, and SES as covariates to account for potential effects. We hypothesize that the initial level of emotion understanding and executive function would predict later peer acceptance and rejection, while the growth rate of emotion understanding and executive function may have different effects on peer relationship over time.

2. Method

2.1. Participants

One hundred and twenty preschool children (70 boys) were invited to participate in the study starting from their first year of kindergarten (Time 1). Children were tested again during their second (Time 2) and third (Time 3) years of kindergarten. In China, kindergarten is the most widely adopted form of full-time early childhood education, catering to children aged three to six. At the age of three, parents often enroll their children in kindergarten for education, which concludes around the age of six when they transition to primary school. These children typically receive three years of kindergarten education, beginning in September and graduating in July after three years. In most kindergartens, children are placed in relatively stable classes with consistent peers and teachers to facilitate their integration into the group and foster a sense of familiarity, except for additional requests from parents to change classes or transfer schools. In this study, the participants were selected from four classes, with class sizes ranging from 25 to 32 children and a fairly balanced distribution of boys and girls within each class. 12 children did not participate in the study at T2 and T3 due to transferring or changing classes, thus, 108 typically developing children (68 boys) without any known learning, language or behavioral dysfunction were included in the analysis. The mean age at Time 1 was 46.63 months ($SD = 3.20$ months, range = 40–54 months), and the mean age at Time 2 was 59.90 months ($SD = 3.22$ months, range = 53–68 months), and the mean age at Time 3 was 70.05 months ($SD = 2.98$ months, range = 65–70 months). Each testing point took place during the second semester of the school year when the students had already become acquainted with their peer group in the class. Before the study, the parents of all children were informed about the purpose of the study and gave their written consent.

2.2. Measures

2.2.1. Emotion understanding: test of emotion comprehension

The Test of Emotion Comprehension (TEC) was used to assess children's emotion understanding at three time points (Pons et al., 2004; Pons et al., 2003). The TEC test involved a picture book depicting simple cartoon scenarios in which a protagonist of the same gender experienced various situations eliciting different emotional responses. The components of TEC were organized in an increasing order of

difficulty (Pons et al., 2004; Pons et al., 2010), thus we only took the first five components of TEC into account because of the age range of the participants in the present study: (1) recognition, (2) external cause, (3) desire, (4) belief, (5) reminder.

The researcher presented a brief statement or story describing a cartoon scenario while simultaneously showing the corresponding image. The children were then asked to identify how the protagonist felt by choosing from four illustrations of faces representing different emotional states (happiness, sadness, fear, or anger). The test consisted of five scenarios, with each scene corresponding to a specific component and presented in a fixed order. These components were: (1) recognizing emotions based on facial expressions (e.g., identifying a happy face), (2) understanding of external causes of emotions (e.g., attributing an appropriate emotion to a protagonist receiving a birthday present); (3) understanding of desire-based emotions (e.g., attributing appropriate emotions to two protagonists facing the same situation but having opposing desires); (4) understanding of emotions based on beliefs (e.g., e.g., attributing an emotion to a sheep who is enjoying green grass without realizing that a wolf is hiding behind the trees); (5) understanding of the influence of a reminder on a present emotional state (e.g., attributing an emotion to a protagonist who is reminded of the loss of a precious object). Additionally, some control questions were included to ascertain the children's precise comprehension of the scenario in the components of the TEC. For instance, in the section pertaining to the understanding of emotions based on beliefs, a control question such as "Who likes cake?" was used, with the correct answer being "The boy likes cake". If children answered incorrectly, the researchers provided correct feedback. Each correct answer earned the children 1 point, resulting in scores ranging from 0 to 5.

2.2.2. Executive function

- (1) **Inhibition Control: Arrows Task (Time 1).** Following the procedure suggested by Davidson and Amos (2006), the arrows task was used to assess children's inhibition control at Time 1. Here, a single large arrow was presented at the left or right of the computer screen. The arrow pointed either straight down or toward the opposite side at a 45 angle. On Congruent trials, the arrow pointed straight down and participants were to respond on the same side as the arrow. On incongruent trials, the arrow pointed diagonally toward the opposite side and participants were to respond on the side opposite the arrow. The precise instructions children were told were, "you need to point in a certain direction according to the arrow. If the arrow is on the side of the box pointing straight down, point the side of arrow. If the arrow is on this side, pointing down across the screen like this, point to the opposite side of arrow". Congruent and incongruent trials were presented in a randomized Mixed block of 20 trials. Stimulus presentation time was 2500ms, and the inter-stimulus interval was 500ms, resulting in total trial duration of 3000ms. The accuracy of the incongruent trials was recorded. The percentage of correct responses (accuracy) was calculated by dividing the number of correct responses by the sum of correct plus incorrect responses. Scores range from 0 to 1.
- (2) **Inhibition Control: Stop-Signal Task (Time 2 and Time 3).** The children's inhibition ability was assessed by a computerized stop-signal task adapted from Logan et al. (1997), which included the go and stop-signal trials. On each trial, children were shown a picture stimulus (flower or heart) in the left side of the screen and a number in the right side of the screen. On 75% of the trials (go-signal trials), they were instructed to press the left arrow button on the keyboard when they saw a flower and to press the right arrow button on the keyboard when they saw a heart, as quickly and accurately as possible. 25% of the remaining trials were stop-signal trials in which a stop-signal number (nine) showed the right side of the screen. At that moment, the participants had to

stop themselves from responding to the target. Trials in which the children successfully inhibited the response were labelled Stop-correct, and trials in which the children erroneously responded to the picture stimulus were labelled commission-error. Children were instructed that stopping and going were equally important. All trials began with a 500ms fixation cross in the center of the screen. Then, the Go stimulus was presented for a 3000ms fixed-response interval. Children were allowed to respond at the start of stimulus presentation until the end of the 3,000ms fixed response interval. Before the formal trials, children practiced one block or 20 trials to ensure that the children understand the rules. The formal task itself consisted of three blocks, each comprising 20 trials. Children were permitted to take brief intermissions between consecutive blocks, allowing them to resume the task when ready. It is worth noting that due to the participants' age range (59 to 70 months), they were unable to make selections by pressing buttons independently. Consequently, the experimenter assisted them in this regard. As a result, only the accuracy rate of responses was recorded, while reaction times were not considered.

In a stop-signal task, two types of errors were primarily observed: commission errors, which occur when a participant responds to a stimulus during stop-signal trials, and omission errors, which occur when a participant fails to respond during go trials (Van de Laar et al., 2014). The percentage of inhibition response failures was calculated by dividing the number of commission errors and omission errors by the total number of responses. Therefore, the accuracy of inhibitory control was represented by 1 minus the percentage of response failures, with scores ranging from 0 to 1.

- (3) **Cognitive Flexibility: Dimensional Change Card Sorting (DCCS).** The dimensional change card sorting (DCCS) task was utilized to assess children's cognitive flexibility at three time points (Zelazo, 2006). In this task, children were presented with two pictures (e.g., a white rabbit and a green boat) and were asked to sort a series of target stimuli (e.g., green rabbits), based on a specific rule (e.g., color). Subsequently, in a post-block, they were instructed to sort the same types of target stimuli according to a different rule (e.g., shape). The two blocks, pre- and post-, were collectively utilized to evaluate the children's proficiency in effectively transitioning between distinct rule sets.

In the experiment phase, each trial began with the simultaneous presentation of two pictures, one showing a white rabbit and the other showing a green boat, followed by the fixation cross along with a rule cue. Subsequently, the target stimuli were presented sequentially at the top of the screen. The children needed to divide the target stimuli according to the rules (color or shape) and make key reactions. A practice phase was conducted to ensure comprehension of the task. We utilized E-prime 2.0 for stimulus presentation and trial response recording. One of the block structure was as follows: The first block (Block 1: color, 6 trials); Block 2 (shape, 6 trials); Block 3 (color, 6 trials); Block 4 (shape, 8 trials); Block 5 (color, 8 trials) (Cepeda & Munakata, 2008). Children were required to switch the rule ('color' to 'shape' or 'shape' to 'color') between adjacent two blocks. We scored as follows: assigned 1 if children passed the first block (Block 1) which used the sorting color rule (children who sort five or more out of six trials correctly). Block 2 switch the sorting rule 'color' to 'shape', and assigned 2 if children passed Block 2 which used the sorting shape rule (children who sort five or more out of six trials correctly); no score was assigned if children failed the block, and so on. The outcome measure used in DCCS was the accuracy rate of switching (Wang et al., 2021; Zelazo, 2006), which represents the ratio of correctly switched trials within each block. The outcome score was calculated by dividing the total score by the number of blocks, with scores ranging from 0 to 1.

- (4) **Working Memory: Memory for Picture of Wechsler Intelligence Scale for Children (IV).** Memory for Picture of Wechsler Intelligence Scale for Children was employed to assess children's working memory at three time points (Wechsler, 2003). In this task, children were required to memorize one or multiple target pictures presented by the experimenter for a fixed duration (3s or 5s) and subsequently identify the target picture from a set of pictures presented later. The working memory test comprised 35 sets of pictures, ranging from one to seven pictures per set. As the number of pictures increased, the difficulty of the task also increased. One point was awarded for each correctly answered question, and the test was concluded when the child failed to answer four consecutive sets of questions. The final score for children's working memory was calculated by summing up the points earned across all sets (Wechsler, 2003).

Based on prior literature, the EF of children was measured using various tasks, including the Arrows Task, the Stop-signal Task, the Dimensional Change Card Sorting (DCCS) task, and the Memory for Picture subtest of the Wechsler Intelligence Scale (Wechsler, 2003; Miyake et al., 2000; Chrysikou & Weisberg, 2005; Zelazo, 2006; Besnier et al., 2008; Li et al., 2020). To ensure equal weighting of the essential components of EF in the study, the raw scores for each core component were first transformed to a 0-1 scale. These transformed scores were then added together to obtain a composite executive function score. The factor structure of EF was examined using confirmatory factor analysis (CFA), and the CFA models demonstrated satisfactory fit indices (Time 1: $\chi^2 = 108.31$, $p < 0.05$, CFI = 0.94, RMSEA = 0.07; Time 2: $\chi^2 = 60.12$, $p < 0.05$, CFI = 0.96, RMSEA = 0.04; Time 3: $\chi^2 = 141.65$, $p < 0.05$, CFI = 0.93, RMSEA = 0.07).

2.2.3. Peer relationships: peer nomination

Social relationship was assessed using the peer nomination sociometric interview technique developed by Coie et al. (1982). This assessment was conducted at Time 2 and Time 3 to gather information about children's relative social status within their peer groups, including peer acceptance and rejection. Peer nominations were collected from the entire class of each participating child. Initially, children were presented with a class photograph and instructed to confirm their familiarity with all the children in the class and to know their names. Subsequently, using photographs of their classmates as prompts, children were asked to nominate three peers they most enjoyed playing with and three peers they least enjoyed playing with. Children received a score of 1 for each nomination received. The following scores were calculated for each child: (a) the number of positive nominations (Peer Acceptance score, PA); (b) the number of negative nominations (Peer Rejection score, PR); (c) a social preference score (SP) based on subtracting the PR score from the PA score; and (d) a social impact score computed by combining the PR score and the PA score. To account for varying class sizes, these scores were standardized within each class. The study maintained the same class and year group composition at Time 3, ensuring that children had consistent access to the same peer group throughout the study, which is an important aspect of the longitudinal design.

Peer nominations have been extensively employed in studies investigating individual variations in children's peer relationships, demonstrating robust test-retest reliability for both peer acceptance ($r = 0.62$, $p < 0.001$) and peer rejection ($r = 0.96$, $p < 0.001$) (Jiang & Cillessen, 2005). As for the relationship between peer acceptance and rejection, no significant association was found across the two time points ($r_1 = -0.19$, $p > 0.05$; $r_2 = -0.17$, $p > 0.05$). This confirms our choice to treat these two variables as distinct entities.

2.2.4. Covariates

To better explore the unique influence of the target variables on the outcomes at the preschool, child age, gender, vocabulary, and family socioeconomic status were included as covariates.

Parent questionnaires were used to gather information about the parents' educational level and family monthly income in order to assess the family socioeconomic status (SES) of the children. Parental education was measured on a 4-point scale: (1) junior high school and below; (2) high school to associate degree; (3) bachelor's degree; (4) master's degree or above. Family monthly household income was assessed on a 4-point scale: (1) less than CNY\$5000; (2) CNY\$5000-10000; (3) CNY\$10000-15000; (4) CNY\$15000 and above. Family SES was calculated by combining the z-scores of family income, mothers' education level, and fathers' education level.

Vocabulary ability was measured using general knowledge of Wechsler Intelligence Scale for Children. The test consists of 4 picture questions and 25 verbal questions, designed to evaluate children's comprehension of general knowledge across various domains, including their vocabulary skills. In the picture question section, children were presented with four pictures and were asked to identify the best answer to the given question asked, for example, "Tell me which one of these can be eaten?" In the verbal question section, children were prompted to answer broad general knowledge questions, such as, "Tell me where your mouth is", "Touch it and show me". One point was awarded for each correct response, and the test concluded when three consecutive questions were answered incorrectly. The maximum total score is 29 points.

2.3. Procedure

Data collection took place during the second trimester of each school year. To align with the kindergarten's teaching schedule, there was a 12-month interval between Time 1 and Time 2, and a 10-month interval between Time 2 and Time 3. At each time point, every child underwent a series of executive function tasks and an emotion understanding task. Children's vocabulary ability and family SES information were collected during Time 1 and used as control variables. Additionally, peer nominations were collected during Time 3, after the children had known each other for least two years.

All tasks were presented in a random order, and children completed the battery of tasks in a session usually lasting between 40 and 70 min. Before the test, the researcher engaged with the children to establish rapport and then conducted the test individually in a quiet room at their school. In the formal test, all of the children understood and complied with the requirements of the task and completed in all sessions. They were allowed a short rest period between tasks until they felt ready to proceed. Their performance was scored by a trained researcher during the task or at the end of the session according to the task procedure and the scoring manuals. All researchers were trained systematically to meet the requirements of testing. The children were rewarded with small gifts such as pencils and toys at the end of each assessment. The entire process will be recorded and taken notes.

2.4. Data analysis plan

Sixteen children were missing data on vocabulary ability, and nine children were missing data on family SES at T1. Missing data analyses indicated that, compared to children with vocabulary data, children missing data scored lower on emotion understanding at T1, $t = 2.21$, $p = 0.07$. Children with missing data did not differ on other variables (i.e., age, gender, executive function, and peer relationships), t s ranged from 0.19 to 1.75, p s ranged from 0.81 to 0.11. Rates of missing data ranged from 0.9% to 3.7% across the target variables at T1 and T2, and these missing were random. Maximum likelihood with robust standard errors (MLR) was applied to address missing data, which performs as well as multiple imputation analyses when data are missing at random (Muthén & Asparouhov, 2002).

Preliminary analyses included descriptive statistics and correlations analyses using Jamovi (version 2.3, url: <https://www.jamovi.org>). The univariate normality of data was evaluated by examining skewness and

kurtosis, with absolute values of skewness lower than 2 and kurtosis lower than 7 considered low departures from normality (Bryne, 2010). None of the variables showed significant departures from normality, thus no transformations were necessary. See Table 2 for details. We employed linear mixed models using 'GAMLj' module in Jamovi to assess whether the change of variables between time points was significant and children's classroom was included as random-effects term to control for their intra-classroom differences.

Main analyses were performed using Latent growth curve model (LGCM) in Mplus Version 8.1 (Muthén & Muthén, 2013). Specifically, LGCM is a type of structural equation model that allows for testing the growth or change in variables based on latent variables (Bollen & Curran, 2006; Karen et al., 2007). The focus of LGCM that makes it distinct from other longitudinal analyses is modeling individual changes by estimating two or more growth terms: initial level of the variable and growth rate per time (in a linear or nonlinear pattern). This approach provides a succinct depiction of the developmental trajectory by estimating growth terms (e.g., intercept and slope) of the growth curve modeling. To gain insight into the temporal dynamics of emotion understanding and executive function, a series of LGCMs were tested. Initially, we fit a linear unconditional model and a quadratic unconditional growth model and then compared their model fits (Little, 2013). Although both linear and quadratic modeling showed good model fit, we chose the linear unconditional growth modeling for its simplicity as the primary model for following analyses. In this linear unconditional modeling, the intercept was a time-independent latent variable, and the factor loading was limited to 1. The slope factor represented the rate of change of EU and EF, with fixed factor loadings of 0, 1, and 2 to indicate time (Bollen & Curran, 2006). Building upon this, we conducted a conditional latent growth model that included peer acceptance and peer rejection as time-invariant dependent variables, examining their relationship with the change in EU and EF (i.e., latent intercept and slope). Additionally, family SES, vocabulary, age, and gender were included as control variables in the model as covariates, as they were only measured at T1. The covariance between EU and EF and the correlation between peer acceptance and rejection were controlled in this model. These analyses allowed us to test the development changes in executive function on peer relationships.

In addition, considering the nesting of children within classrooms, a multi-group analysis was used to test the model for the presence of classroom-level effect. We excluded data from one class (e.g., class1) and formed a group of remaining data for comparison with the larger group (including all data). Firstly, all structural coefficients were constrained to be equal across the groups to obtain the fully constrained model. Secondly, another unconstrained model was estimated, in which all the parameters except for factor loading whose values were constrained to equality were freely estimated across the groups. Model comparison results showed that there was no significant difference between the fit of the fully constrained model and the unconstrained model ($\Delta\chi^2 = 3.02$, $\Delta df = 2$, $p = 0.42$). Such multi-group analyses were conducted four times. Thus, the classroom fixed effects due to classroom level could be excluded, and the analysis was also used as a robustness check to strengthen the modeling effect.

To investigate the key role of acceptance and rejection on peer relationships, we calculated the total index of social preference as the dependent variable and conducted an additional conditional model. Model fit for the model was evaluated, but the fit indexes of the conditional model was not acceptable, $\chi^2/df = 46.09/21$, RMSEA = 0.119, CFI = 0.883, TLI = 0.805, SRMR = 0.106. Therefore, this further confirmed that the two indicators of peer acceptance and peer rejection were the most appropriate ones to measure children's peer relationships.

3. Results

3.1. Descriptive statistic and preliminary analysis

Table 1 presented means, standard deviations, and correlations among emotion understanding, executive function and covariates. Correlation analyses showed that peer acceptance was positively correlated with emotion understanding and executive function at three time points, whereas peer rejection was negatively correlated with executive function at three time points, and was negatively correlated with emotion understanding at T1 and T3. However, gender, family SES, and vocabulary were correlated with the variables of interest. As such, we controlled for children's age, gender, family SES and vocabulary in the subsequent analysis.

To explore the changes in emotion understanding and executive function, we conducted a mixed repeated measures analysis. Time was treated as a fixed-effects factor, while classrooms were included as a random factor to account for the effect of classroom-level. We found significant differences among the three measures of emotion understanding ($F(1, 104) = 37.1, p < 0.001$), and executive function ($F(1, 104) = 23.08, p < 0.001$). That is, the means for emotion understanding and executive function increased significantly over time (see Fig. 1). The inter-class correlations from random effect were 0.12 and 0.10, indicating that classroom-level accounted only for 10–12% of variance.

3.2. Unconditional latent growth model (LGM)

We employed a the linear unconditional growth model to examine the temporal dynamics of children's emotion understanding and executive function. Table 2 presented the model fit indices, intercept, and slope of the liner unconditional LGM. Both the liner models of emotion understanding and executive function showed a good fit. The results revealed a significant linear increase over time in emotion understanding ($\beta = 0.82 (0.10), p < 0.001$), and executive function ($\beta = 0.59 (0.06), p < 0.001$). There was also a significant variance of the intercept, indicating that there were significant differences among children in the initial emotion understanding ($\sigma^2 = 2.44 (0.52), p < 0.001$), and executive function ($\sigma^2 = 1.53 (0.25), p < 0.001$). Moreover, the intercept and slope were negatively correlated in emotion understanding ($r = -0.37, p < 0.05$), and executive function ($r = -0.58, p < 0.001$), indicating that higher initial levels were associated with slower growth rates. Based on the unconditional LGCM, we further explored the relationship of peer relationships with the developmental trajectory of emotion understanding and executive function.

Table 1

Descriptive statistics of and correlations among the primary variables and covariates.

Measure	1	2	3	4	5	6	7	8	9	10	11	12
1.Age	-											
2.Gender	.06	-										
3.Family SES	.10	.12	-									
4.Vocabulary	.11	.04	.09	-								
5.EU_T1	.07	-.02	.12	.31**	-							
6.EU_T2	-.05	.07	.18*	.27*	.44**	-						
7.EU_T3	.11	.13	.17	.20*	.39**	.42**	-					
8.EF_T1	.08	.10	.12	.24*	.34**	.33**	.30**	-				
9.EF_T2	.10	.04	.11	.19*	.34**	.30**	.35**	.48**	-			
10.EF_T3	.12	.04	.19*	.14	.29**	.25*	.38*	.31*	.44**	-		
11.PA_T3	.05	.10	.13	.22*	.31**	.22*	.29*	.32**	.26*	.24*	-	
12.PR_T3	.07	-.28*	-.12	-.08	-.23*	-.19	-.30*	-.24*	-.48**	-.42**	-.16	-
Mean	46.63	.63	.01	14.88	2.29	3.48	4.07	1.60	2.05	2.67	.02	-.06
SD	3.20	.50	.87	7.59	.91	.86	.73	.66	.43	.54	.46	.52
Skewness	.35	.34	.35	.13	-.17	-.66	-.83	-.65	-.65	-.14	1.08	1.23
Kurtosis	-.58	-1.90	-.82	.73	-.65	.14	.18	.18	-.30	-.87	1.27	2.04

Note. N=108.T1, Time1; T2, Time2; Time3. EU, Emotion Understanding; EF, Executive Function; PA, Peer Acceptance; PR, Peer Rejection. The coding for male is 1, and the female is 0.

* $p < 0.05$. ** $p < 0.01$.

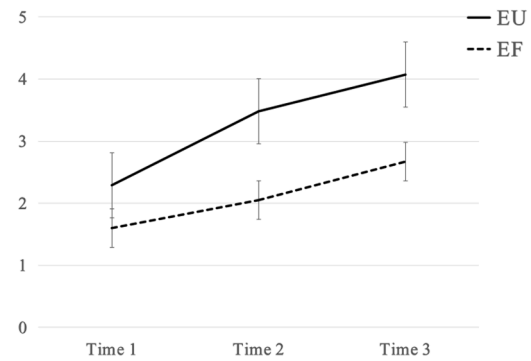


Fig. 1. The change of executive function and emotion understanding at three time points

Note. EU, Emotion Understanding; EF, Executive Function.

3.3. The conditional latent growth model

To test whether the initial level and growth rate of emotion understanding and executive function can predict children's peer relationships, the conditional latent growth model was analyzed. Specifically, we included peer acceptance and peer rejection at T3 in the model as dependent variables to examine how they relate to the development of emotion understanding and executive function (i.e., latent intercept and slope), controlling for the covariates of age, gender, family SES and vocabulary at T1. The results of the conditional LGCM with all paths are presented in Fig. 2, and the model fit is deemed acceptable (see Table 2). Notably, several covariates exhibited non-significant associations with peer relationships. Similarly, non-significant connections were identified between emotion understanding and peer rejection. Following Little's (2013) recommendations, we pruned non-significant paths to generate more parsimonious models and to increase statistical power. The marginally significant path from emotion understanding to peer acceptance ($\beta = 0.08, p = 0.095$) and from executive function to peer acceptance ($\beta = 0.11, p = 0.072$) were preserved. The final model with all paths is presented in Fig. 3, and it had a good fit to the data, $\chi^2/df = 54/15$, RMSEA = 0.032, CFI = 0.971, TLI = 0.979, SRMR = 0.046.

As showed in Table 3, the intercept of emotion understanding can significantly and positively predicted peer acceptance ($\beta = 0.14 (0.03), p < 0.05$), as did the intercept of executive function ($\beta = 0.18 (0.03), p < 0.05$). These findings indicated that children with higher initial level of emotion understanding and executive function were more likely to be accepted by their peers. The intercept and the slope of executive

Table 2
Model fit of unconditional latent growth model and conditional latent growth model.

Model	χ^2/df	RMSEA	CFI	TLI	SRMR	Mean(variance)		<i>r</i>
						Intercept	Slope	
EU_LGCM	6.271	0.033	0.929	0.948	0.071	2.44(0.52)**	0.82(0.10)**	-0.37*
EF_LGCM	4.041	0.010	0.931	0.979	0.058	1.53(0.25)**	0.59(0.06)**	-0.58**
Conditional LGCM	10.33	0.071	0.909	0.921	0.052			
Final LGCM	3.60	0.032	0.971	0.979	0.046			

Note. EU, Emotion Understanding; EF, Executive Function; LGCM, Latent Growth Curve Model; RMSEA, Root Mean Square Error of Approximation; CFI, Comparative Fit index; TLI, TuckerLewis index; SRMR, Standardized Root Mean Square Residual; Intercept mean initial status, Slope mean rate of change; *r* mean correlation between initial status and rate of change.

* $p < 0.05$. ** $p < 0.01$.

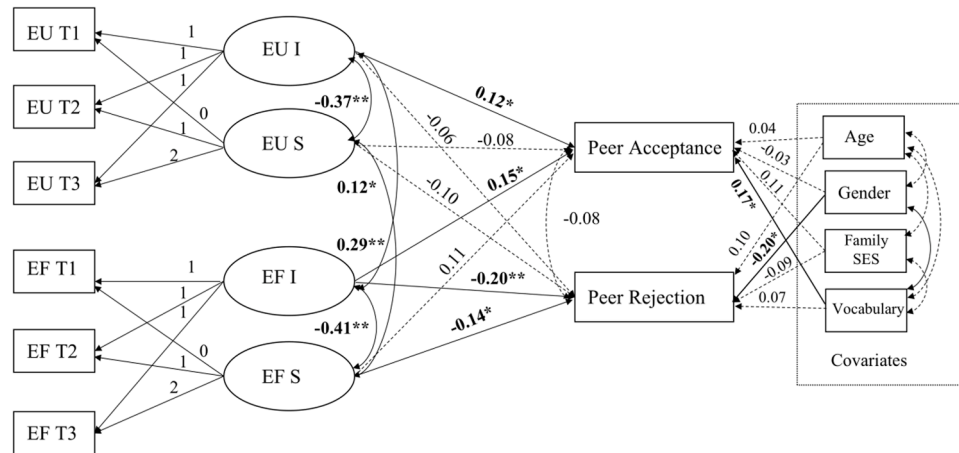


Fig. 2. The conditional latent growth model with peer relationships as dependent variables.

Note. The standardized coefficients are reported. Full line are significant ($p < 0.05$). Dotted values are not significant ($p > 0.05$). Error terms and error covariance terms are not shown. EU, Emotion Understanding; EF, Executive Function; I, Intercept; S, Slope; T1, Time 1; T2, Time 2; T3, Time 3. * $p < 0.05$; ** $p < 0.01$.

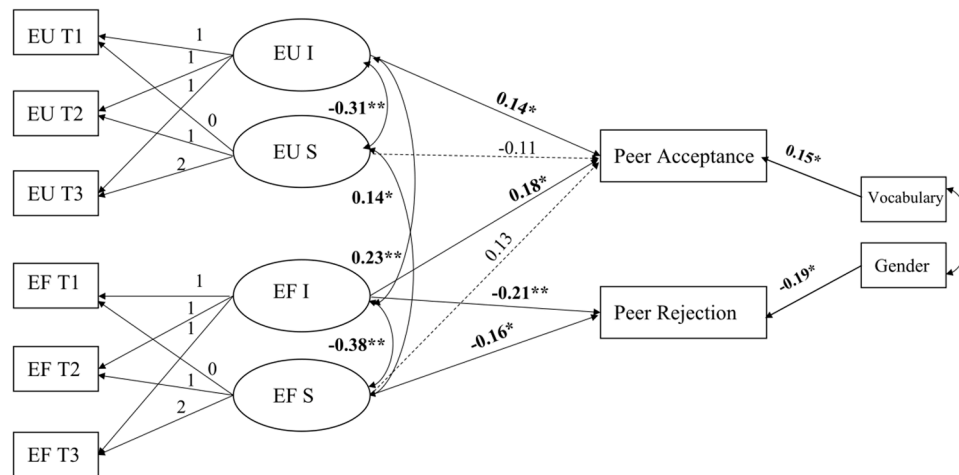


Fig. 3. The final conditional model with all significant paths.

Note. The non-significant path from EU S to Peer Acceptance and EF S to Peer Acceptance were preserved. The standardized coefficients are reported. Error terms and error covariance terms are not shown. EU, Emotion Understanding; EF, Executive Function; I, Intercept; S, Slope; T1, Time 1; T2, Time 2; T3, Time 3. * $p < 0.05$; ** $p < 0.01$.

function could significantly and negatively predict peer rejection ($\beta = -0.21$ (0.02), $p < 0.05$, and $\beta = -0.16$ (0.11), $p < 0.05$), while the intercept and the slope of emotion understanding had no significant effect on peer rejection ($\beta = -0.06$ (0.17), $p = 0.71$, and $\beta = -0.10$ (0.13), $p = 0.36$), suggesting that children with higher initial level and faster growth rate of executive function are less likely to be rejected by peer groups, but emotion understanding does not contribute significantly to

peer rejection. Thus, the initial levels of emotion understanding and executive function have significant positive effects on the formation of positive peer relationships later in young children, especially in obtaining more peer acceptance. Furthermore, the growth rate of executive function was also significantly related to peer relationships, as evidenced by the prediction of lower levels of peer rejection. Analyzing the number of significant pathways in Fig. 3, we found that only one

Table 3
Standardized path coefficients for the conditional latent growth model.

path	β	SE	p
Variable predicting the peer acceptance			
The Intercept of EU	0.14	0.03	0.039
The Slope of EU	-0.11	0.17	0.087
The Intercept of EF	0.18	0.03	0.025
The Slope of EF	0.13	0.10	0.070
Variable predicting the peer rejection			
The Intercept of EF	-0.21	0.02	0.001
The Slope of EF	-0.16	0.11	0.031

pathway linking emotion understanding to peer relationships was significant, with a path coefficient of only 0.14, whereas executive function had three significant pathways with path coefficients greater than that of emotion understanding. These results suggest that executive function plays a more prominent role than emotion understanding in fostering the formation of positive peer relationships in preschool children.

4. Discussion

This study employed a conditional latent growth model to examine the different effects of the initial level and growth rate of emotion understanding and executive function on peer relationships. As hypothesized, the initial levels of emotion understanding and executive function positively predicted later children's peer acceptance, and the initial level of executive function negatively predicted peer rejection, even after controlling for age, gender, family SES, and vocabulary. Moreover, the latent growth modeling analysis revealed that children with faster growth rate in executive function were less likely to be rejected by peers and may gain more peer acceptance, but the development of emotion understanding had no effect on peer relationships. These findings indicate that executive function can facilitate positive peer relationships more than emotion understanding in preschool children, which provides empirical evidence for understanding the role of executive function in peer relationships from a temporal dynamics perspective.

4.1. Emotion understanding, executive function and peer acceptance

Our findings demonstrated that the initial level of emotion understanding and executive function positively predicted later peer acceptance, with controlling for other covariates. That is, when forming new relationships, children who exhibit control behaviors, solve problems and understanding others' emotions were more likely to be rated positively by their peers. These findings align with the prevailing consensus that both executive function and emotion understanding play vital roles in shaping children's social relationships (Slaughter et al., 2015; Hendry et al., 2016; Caporaso et al., 2019; Moriguchi, 2014; O'Toole et al., 2017). Firstly, children with more advanced level of emotion understanding may be at an advantage when establishing peer relationships due to their heightened ability to sense their own emotions as well as understand the perspectives of their peers. In other words, emotion understanding allows children to read and respond to complex social situations more effectively (Banerjee, 2011; Malti & Noam, 2016). Secondly, successful peer interactions require friendlier and more self-regulated behaviors (Auther et al., 2018; Ladd, 2005; Denham & Brown, 2010). Thus, children with a high level of executive function are better equipped to adapt to social interactions and solve problems more effectively. These factors contribute significantly to the robust relationship observed between emotion understanding and executive function with children's peer acceptance. Prior research has identified various aspects of children's abilities and skills (e.g., emotion understanding, inhibition control, working memory), as discrete factors, influencing peer relationships (Ocampo et al., 2020). However, this represents a limitation in our comprehensive comprehension of peer relationships. Our findings are particularly promising as they encompass

the measurement of both emotion understanding and executive function in a single study, shedding light on their concurrent impact on peer acceptance.

4.2. Effect of the growth rate of executive function on peer relationships

Latent growth modeling showed that only executive function negatively predicted later peer rejection, and children who showed a faster growth rate in executive function were less likely to experience rejection from their peers. Our study suggests that the benefits of advanced executive function for children's peer relationships can be extended to middle childhood, especially with the growth rate of executive function playing the key role. Executive function includes the ability to retain and manipulate information in the mind (working memory), to inhibit inappropriate responses (inhibition control), and to adapt to changing environments (cognitive flexibility) (Davidson et al., 2006; Hendry et al., 2016; Miyake & Friedman, 2012). The close association between executive function and peer relationships in childhood has been emphasized by these functions (Lecce et al., 2020). The social model has emphasized the role of executive function in social interaction (Beauchamp & Anderson, 2010). According to this model, executive function permits one to inhibit impulses reactions and adapt to changes in the ongoing social environment, facilitating the coordination of goals and behavior during social interactions. Our study supports this account by demonstrating that preschool children with higher executive function are more likely to be accepted by peers and less likely to experience rejection, which also is consistent with previous studies (Holmes et al., 2016; Garcia-Andres et al., 2010). Importantly, the cooperative games among peers gradually increase over time with decreasing the number of solo and parallel games, which make the opportunities for problems-solving and managing conflicts within peers begin to increase (Sterba et al., 2007; Parker et al., 2015; Thorsen et al., 2018; Mendez & Fogle, 2002). Concurrently, children's executive function begins to play a more prominent role in controlling their behaviors and thoughts within peer relationships. Therefore, the growth rate of executive function can have an impact on children's peer rejection over time. This study fills a gap in our understanding of how the development of executive function affects the formation of peer relationships during the preschool years, and emphasizes the crucial role of executive function in social interactions.

4.3. Effect of the growth rate of emotion understanding on peer relationships

In contrast, the current study indicated that the effect of emotion understanding on the formation of peer relationships was relatively weaker. Our data showed that only the initial level of emotion understanding had a significant impact on later peer acceptance, while other paths were not statistically significant. Previous views have implied that the effect of emotion skills, including emotion understanding on social relations should be strongest when children enter a new environment (Brownell & Hazen, 2010), and once children have established themselves in a new peer group, this effect becomes relatively stable or even weaken (Fink et al., 2014). Thus, the initial level of emotion understanding determines the acceptance degree of children when entering new peer groups, while the growth rate has less contribution to peer relationships over time. As peer groups become more familiar and children grow older, they become involved in larger social groups and engage in more complex forms of social activities, including conversations, playing games with rules, cooperating/competing with peers, and problem-solving (Andrew, 2020; Rubin et al., 2006). Two aspects of preschool children's social life (namely age groups and complex interactions) are closely related to executive function, as they require the ability to retain and manipulate information in the mind, inhibit inappropriate responses, and adapt to a changing environment. In other words, although the establishment of peer relationship in younger

children is significantly affected by emotion understanding, the maintenance of peer relationship in childhood may depend on other abilities, such as inhibition control, cognitive flexibility, and working memory. Therefore, emotion skills are not the primary indicators of children's peer selection. Instead, the ability to engage in more challenging games and problem-solving becomes more important for children to maintain and stabilize peer relationships (Harms et al., 2014; Holmes et al., 2016; Moriguchi, 2014). Although emotion understanding undeniably contributes to greater peer acceptance when entering new groups, executive function may further facilitate the establishment and maintenance of positive relationships. It achieves this by reducing behavior problems and enabling a diverse range of conflict resolution strategies, thereby reducing peer rejection (McQuade et al., 2013; Lecce et al., 2020). This intriguing possibility highlights the dynamic process between emotion understanding and executive function in the formation of social interactions.

4.4. The indicators of peer relationships

It is important to note that the current study employed peer acceptance and rejection as indicators of children's peer relationships within groups. This offers unique insights into how children perceive and integrate their social relationships among peers at school (Fink et al., 2014; Wang et al., 2021). These insights are not typically captured by measures of social relationships obtained from teachers or parents (Babad, 2001; Cillessen & Mayeux, 2004). Because the extent to which children like and dislike their peers serves to structure children's groups and to determine the place of individual children in the group, therefore, exploring peer relationships from the perspectives of acceptance and rejection can provide a more comprehensive and detailed understanding of children's social standing and interpersonal dynamics. In addition, there were no significant correlation between peer acceptance and peer rejection at two time points ($r_1 = -0.19, p > 0.05$; $r_2 = -0.17, p > 0.05$), which confirmed previous findings that acceptance and rejection are not the toe endpoints of peer relationships and represent different aspects of peer relationships (Bukowski et al., 2010). This further supports the meaningfulness of considering peer acceptance and rejection as indicators of peer relationships.

4.5. Implications in the Chinese educational settings

In addition, the distinctive characteristics of the Chinese educational settings could provide a cultural interpretation for the findings of this study. In Chinese society, as well as in other collectivist societies, individuals are encouraged to regulate their desires and actions in order to foster social harmony and facilitate group cohesion (Chen, 2010; Greenfield et al., 2006). This cultural background may have implications for the development of executive function, especially when children with stronger executive function receive social approval that helps them display strengths and form positive peer relationships. Unlike in other educational cultures, being "smarter" tends to be highly valued by parents and teachers in China, while being "more understanding" may not be given as much importance. For example, when Chinese preschoolers were asked to rate which children were least popular, the reasons most frequently mentioned were being not smart and disliked by parents/teachers (Ding et al., 2014). At the same time, parents also create opportunities for their children to spend time with "smarter" peers, such as saying hello actively on the road and mentioning it repeatedly at home. This creates an environment in which children with stronger executive function ("smarter") may not encounter difficulties in peer interactions, further emphasizing why executive functioning is more critical in peer relationship formation.

The transition to school is a critical stage in a child's social and developmental growth, as they begin to engage in social interactions and form new relationships, particularly with their peers (Engdahl, 2012; Ortega et al., 2009; Rubin et al., 2006). It is therefore of practical

importance to clarify the relation between cognitive skills and children's social relationships. The findings of this study imply that if we want to make the most impact on social relationships, we should support children's early understanding of emotion and improve the level of executive function when they enter school. In fact, there has been a rise in social skills interventions in preschool settings in recent years, specifically designed to foster positive peer relationships among children (Slaughter et al., 2015; Hamre et al., 2012). However, the social challenges and opportunities facing children cannot be oversimplified, and emotion understanding only provides a singular explanation for children's social interaction. Alongside fostering emotion understanding, it is imperative that intervention training also focuses on enhancing fundamental cognitive abilities (Diamond & Lee, 2011; Diamond & Ling, 2016), such as executive function. Thus, parents and educators should offer educational activities to help children who have social difficulties to improve their peer relationships by enhancing their executive function. This longitudinal study provides a unique insight into the role played by executive function in the formation of peer relationships and highlights the lasting power of peer group dynamics, and by doing so provides practical implications for the timing and design of interventions to promote positive peer relationships.

4.6. Limitations and future research directions

It is important to acknowledge the limitations of the current study and the ensuing potential directions for future research. The first limitation pertains to the sample size. Conducting a detailed 2-year longitudinal investigation on social-cognitive development among 120 young children is indeed a commendable accomplishment. Nonetheless, a larger sample size would significantly enhance statistical power and potentially unveil relationships that did not achieve significance in the present study. For instance, it is entirely plausible that early emotion understanding could impact children's experience of peer rejection, as observed in recent studies involving older children (Fonseca et al., 2009). Secondly, in addition to emotion understanding and executive function, there are numerous other ability (such as theory of mind, prosocial behavior and academic functioning) that have been shown to be associated with social relationships at groups. Due to the size constraints of the current study, these factors could not be considered (for a review see Rubin et al., 2006). Hence, future research should aim to integrate multiple specific cognitive and social cognitive abilities and systematically explore their effects and underlying mechanisms on children's peer relationships. Furthermore, it should be noted that some of the specific findings may not be directly comparable to Western studies due to the virtual absence of research on these issues. For instance, the significant impact of executive function observed in this study may be influenced by the emphasis on academic achievement within the Chinese group-oriented context. In Western societies where early understanding is promoted, emotion understanding may play a more prominent role in peer relationships. Thus, this study needs to be replicated in other cultural contexts. Lastly, this study solely examined the impact of executive function and emotion understanding on the formation and stability of peer relationships through a time dynamics perspective, without delving into the specific influence mechanisms. Future studies can further explore and refine the nature of the relationship between these constructs.

5. Conclusion

This study systematically examined the developmental effects of emotion understanding and executive function on specific dimensions of peer relationships (acceptance and rejection) in preschool children through a latent growth model. Our results indicated that a higher initial level and faster growth rate of executive function exert a more pronounced effect in promoting positive peer relationships in preschool children, when compared to emotion understanding. Given that poor

social relations during the early childhood have been shown to have far-reaching implications for later relationships and adjustment, this study offers a unique contribution to our understanding of the longitudinal processes that underlie children's social relationships. Furthermore, it provides valuable insights for interventions designed to support children at risk of experiencing social isolation or peer group rejection.

CRedit authorship contribution statement

Xueke Wang: Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Tingyong Feng:** Conceptualization, Methodology, Supervision, Funding acquisition, Investigation, Writing – review & editing.

Declaration of Competing Interest

All authors declare that they have no conflict of interests.

Data availability

Data will be made available on request.

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